What the Second Congruence Buys: Quantified Evidence on Popovych's Conjecture

(standalone note — v0.1)

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Audit revision (22 October 2026). The support lower bounds below are conditional, not consequences of the collapse upper bounds alone. Special finite-field collapses are possible. The same-name PDF beside this source is historical. The rebuilt audit version is in `doc/audit-pdfs/`; see `doc/integration-review-status.md`.

Abstract

Agrawal's conjecture asserts that for a prime $r \nmid n$ the congruence $(X-1)^n \equiv X^n - 1 \pmod{n, X^r-1}$ forces n prime or $n^2 \equiv 1 \pmod r$; Popovych (2009) conjectured that adding the second congruence $(X+2)^n \equiv X^n + 2$ repairs the statement. We compare support models for Frobenius-matching constructions. The first section has the exact inversion identity $(X-1)^{p^{d/2}-1} = -X^{-1}$ for even $d = \mathrm{ord}_r(p)$. For the second section, a characteristic-zero ideal argument proves that $(\zeta_r^\gamma + 2)/(\zeta_r + 2)$ is not a unit for $\gamma \neq 1$. This excludes a global root-of-unity identity, not exceptional finite-field collapses or missing order factors.

Under explicit section-specific support-sharpness hypotheses at a fixed smoothness threshold, the pair's containing-torus dimension is $\min\{d : d \mid r-1,\ d \geq 3\}$, versus the single-section model's minimum 2. At $r = 5$ the pair model has dimension 4, requiring smooth $p-1$, $p+1$ and p^2+1. The lower smoothness requirements are *conditional*; upper order bounds alone do not prove them. Numerical order checks and parameter-specific Dickman estimates are finite evidence and heuristics, not a proof that every feasible harvest is empty. We make no priority claim, proof of either conjecture, or general algorithmic impossibility claim.

1 Introduction

For coprime n, r write $T(a, n, r)$ for the congruence $(X+a)^n \equiv X^n + a \pmod{n,\ X^r-1}$. Agrawal's conjecture [1, 2] states that $T(-1, n, r)$ with r prime, $r \nmid n$, forces n prime or $n^2 \equiv 1 \pmod r$; it is open, and believed false on the strength of the Lenstra–Pomerance (L–P) heuristic construction [3]. Popovych [4] proved a generalization of the L–P counterexample mechanism and proposed the repaired conjecture:

Conjecture 1 (Popovych)**.** *If r is a prime not dividing n and both $T(-1, n, r)$ and $T(2, n, r)$ hold, then n is prime or $n^2 \equiv 1 \pmod r$.*

Its evidence base is thin in both directions: a plausibility argument that the pair $\{X-1, X+2\}$ generates a large unit subgroup [4, 5], and the Primaboinca exhaustive search ($10^{10} < n < 10^{17}$, nothing found). This note provides a conditional comparison of containing support products: global model minima 3 and 2, and dimensions 4 and 2 at $r = 5$. At some other fixed moduli the two models tie; a larger global minimum is not a strict increase in every cell.

We consider the *Frobenius-matching* family containing the original L–P sufficient construction [3]. In the Φ_r factor modulo each prime $p \mid n$, matching holds if $n \equiv p^j \pmod{\operatorname{lcm}(r, \operatorname{ord}(X+a))}$, where the order is taken in $(\mathbb{F}_p[X]/\Phi_r(X))^\times$, because $(X+a)^{p^j} \equiv X^{p^j} + a$ exactly (Frobenius and Fermat). A construction assembles a squarefree $n = p_1 \cdots p_k$ meeting these congruences at every p_i simultaneously. Requiring those orders to be B-smooth is the particular assembly design analyzed here, not a necessary condition for every conceivable construction. Within that design, the support is where the two sections differ. On the $X = 1$ factor one must also impose $(1+a)^n \equiv 1 + a \pmod{p}$; matching in the Φ_r factor alone does not imply this. It is automatic for a squarefree Carmichael number, including when the base is zero modulo a factor.

2 The collapse for $X - 1$ and its absence for $X + 2$

Fix an odd prime $r \neq 3$, a prime $p \nmid 3r(2^r + 1)$, and let $d = \operatorname{ord}_r(p)$. Write $\rho_r(p)$ and $\rho'_r(p)$ for the multiplicative orders of $X - 1$ and $X + 2$ in $(\mathbb{F}_p[X]/\Phi_r(X))^\times$; both divide $p^d - 1 = \prod_{e \mid d} \Phi_e(p)$, a factorization indexed by *cyclotomic levels* e, with $\Phi_e(p)$ the order of the e-th cyclotomic torus over $\mathbb{F}_p$, of dimension $\varphi(e)$.

Lemma 2 (inversion collapse for the first section)**.** *If d is even, then $(X-1)^{p^{d/2}-1} = -X^{-1}$ in $\mathbb{F}_p[X]/\Phi_r(X)$, and hence $\rho_r(p) \mid 2r(p^{d/2} - 1) = 2r \prod_{e \mid d/2} \Phi_e(p)$.*

Proof. $(\mathbb{Z}/r)^\times$ is cyclic, so its unique involution is -1 and $p^{d/2} \equiv -1 \pmod{r}$. In each field component, with $X \mapsto \zeta$ a primitive r-th root of unity, iterated Frobenius gives $(\zeta - 1)^{p^{d/2}} = \zeta^{-1} - 1 = -\zeta^{-1}(\zeta - 1)$; divide by $\zeta - 1 \neq 0$. Finally $(-X^{-1})^{2r} = 1$ since $X^r \equiv 1 \bmod \Phi_r$. □

For $r = 5$, $d = 4$ this is Váňa's $\rho_5(p) \mid 10(p^2 - 1)$ [6], the fact that makes the L–P construction's cost the *double* condition ($p - 1$ and $p + 1$ smooth: dimension $1 + 1 = 2$) rather than the full $p^4 - 1$. The collapse exists because the ratio under inversion is torsion, with root-of-unity cofactor. For the second section the global conjugate ratio has a different property:

Lemma 3 (trivial stabilizer)**.** *For every $\gamma \in (\mathbb{Z}/r)^\times$ with $\gamma \neq 1$, the element $(\zeta^\gamma + 2)/(\zeta + 2)$ is not a unit of $\mathbb{Z}[\zeta_r]$ — in particular not a root of unity. There is therefore no characteristic-zero torsion-cofactor identity for these conjugate ratios. This does not forbid accidental finite-field relations of the form $(X+2)^{p^e-1} \in \mu_{2r}$.*

Proof. First, $N(\zeta + 2) = \Phi_r(-2) = (2^r + 1)/3$ (from $X^r - 1 = (X-1)\Phi_r(X)$ at $X = -2$), which is coprime to r: by Fermat $2^r \equiv 2 \pmod{r}$, so $2^r + 1 \equiv 3$, and $\gcd(r, 3) = 1$. Let $\mathfrak{q}$ be a prime of $\mathbb{Z}[\zeta_r]$ dividing $(\zeta + 2)$; then $\zeta \equiv -2 \pmod{\mathfrak{q}}$, and the image of -2 in the residue field has multiplicative order exactly r (it is the image of ζ). If $\mathfrak{q}$ also divided $(\zeta^\gamma + 2)$ then $\zeta^\gamma \equiv -2 \equiv \zeta$, so $\zeta^{\gamma - 1} \equiv 1 \pmod{\mathfrak{q}}$, forcing $r \mid \gamma - 1$. Hence for $\gamma \neq 1$ the ratio has a pole at $\mathfrak{q}$ and is not a unit. □

Remark 4 (a special reduction). For $r = 13$ and $p = 5$, one has $5^2 \not\equiv 1 \pmod{13}$ but $(X+2)^{24} \in \mu_{26}$ in $\mathbb{F}_5[X]/\Phi_{13}(X)$. This permitted prime illustrates why the global lemma is not a no-exception statement about reductions.

Remark 5 (two structural asides). (i) The $X = 1$ component of $T(2, n, r)$ is the base-3 Fermat condition $3^n \equiv 3 \pmod{n}$ — automatic for Carmichael-type n, but nontrivial, unlike the first section's $0 = 0$. (ii) Modulo $p = 3$ one has $-2 \equiv 1$, so the two sections *coincide*: the pair constrains only the divisors $p > 3$ of n. Finitely many p (those dividing $2^r + 1$) make $X + 2$ a zero divisor and are likewise excluded.

3 The burden jump

Definition 6 (sharpness hypothesis)**.** *Fix $B \geq 2rd$. For the chosen section f and containing product $P_f(p)$, assume that every prime $\ell > B$ dividing $P_f(p)$ also divides* $\mathrm{ord}(f)$*. For $X-1$ take $P_f = p^{d'} - 1$, where $d' = d/2$ for even d and $d' = d$ otherwise; for $X+2$ take $P_f = p^d - 1$. These are separate assumptions about prime support, not full prime-power divisibility.*

In a cyclic group of order N with $a = v_\ell(N)$, missing ℓ from an element's order is equivalent to being an ℓ^a-th power, not merely an ℓ-th power when $a > 1$. A uniform element misses it with probability ℓ^{-a}; independence and uniformity for the actual conjugate values have not been established.

Theorem 7 (Popovych burden)**.** *Let n be produced by the Frobenius-matching mechanism for the pair: for every $p \mid n$ there is j_p with $n \equiv p^{j_p} \pmod{\mathrm{lcm}(r, \rho_r(p), \rho'_r(p))}$, and $\mathrm{ord}_r(n) > 2$ (validity). Assume Definition 6 for both sections at a common $B \geq \max_{p|n} 2rd_p$. In the design requiring all matching orders to be B-smooth, for every $p \mid n$, writing $d_p = \mathrm{ord}_r(p)$ (and noting validity forces $d_p \geq 3$), the quantity $\prod_{e|d_p} \Phi_e(p)$ — a product of cyclotomic torus orders of total dimension $\sum_{e|d_p} \varphi(e) = d_p$ — must be B-smooth. Prime support alone does not imply that this whole product divides the common modulus with its full multiplicities. In particular the minimal per-prime burden over admissible cells is*

$$\mathrm{burden}_{\mathrm{Pop}}(r) \;=\; \min\{\, d : d \mid r-1,\; d \geq 3 \,\} \;\geq\; 3 \qquad \textit{for every prime } r,$$

whereas the single-section (Agrawal) mechanism has burden $\min_{d|r-1,\, d\geq 3}(d/2$ if d even, else $d)$, with global optimum 2. At $r = 5$: $\mathrm{burden}_{\mathrm{Pop}} = 4$, i.e. $p-1$, $p+1$ and p^2+1 simultaneously smooth, against the single section's 2.

Proof. For $X+2$, its order divides $p^{d_p} - 1$, so smoothness of that product is sufficient. Under Definition 6 every prime above B in the product must also divide the section order, proving necessity. This uses the support hypothesis, not Lemma 3 alone. The containing product has dimension d_p for both parities. Minimizing d_p over $d \mid r-1$, $d \geq 3$ gives the formula; the least such divisor is 3 iff $3 \mid r-1$, and for $r = 5$ (where $r-1 = 4$) it is 4, with levels $\{1, 2, 4\}$ and $\Phi_4(p) = p^2 + 1$. The Agrawal comparison is Lemma 2 plus the same minimization. □

Remark 8 (heuristic cost at representative parameters). For pool primes of b bits and smoothness bound B, the extra quadratic level of Theorem 7 at $r = 5$ costs a Dickman factor $\rho(2b/\log_2 B)$ beyond the double condition. At the parameters of the Agrawal-directed harvest study in this repository ($b = 51$; $B = 1259$ for the solver base, $B' = 12590$ for the enumerated base), this is $\rho(9.9) \approx 3.9 \times 10^{-11}$, respectively $\rho(7.5) \approx 1.8 \times 10^{-7}$ — reducing that study's projected pool of 2.5×10^5 elements to an expected size between 10^{-5} and 0.05. This assumes suitable polynomial-value smoothness and conditional independence. It is neither a proof of emptiness at every feasible scale nor an exclusion of other structured harvests.

r	Agrawal burden	Popovych burden	Δ
5	2	4	+2
7	3	3	0
11	5	5	0
13	2	3	+1
17	2	4	+2
19	3	3	0
23	11	11	0
29	2	4	+2
31	3	3	0
37	2	3	+1
41	2	4	+2

The deltas vanish exactly at $r \equiv 3 \pmod 4$, where the Agrawal burden is already ≥ 3 and both conjectures share full depth; the Agrawal global optimum (2, at any $r \equiv 1 \bmod 4$) is unavailable to the pair at every modulus.

Remark 9 (the constant 2 is immaterial). For any integer $c \notin \{0, \pm 1\}$, the characteristic-zero argument of Lemma 3 applies with the exclusions below. The conditional calculation of Theorem 7 additionally requires its own support-sharpness hypothesis for the pair $\{X-1, X+c\}$: the proof needs only that at a prime $\mathfrak{q} \mid (\zeta + c)$ the image of $-c$ has multiplicative order r, which holds at the primitive prime divisors of $N(\zeta + c) = \Phi_r(-c) = (c^r + 1)/(c+1)$. The constant enters only through that norm (its primes, with $p \mid c(c+1)$, are the finitely many excluded p), the excluded modulus $r \mid c + 1$ (for $c = 2$: our standing $r \neq 3$), and the level-one component, the base-$(c{+}1)$ Fermat condition of Remark 5. What must be avoided is exactly degeneracy of the zero $-c$: it must be a point of $\mathbb{G}_m$ avoiding the rational torsion $\{\pm 1\}$, i.e. $c \notin \{0, \pm 1\}$. The exclusion $c = 0$ is vacuous rather than subtle: X is a monomial, so $(X)^n \equiv X^n$ is an identity in every ring (the zero sits at the origin, off the torus, and the section is a unit of finite order r). Among nonzero c: $c = -1$ is the first section itself, and $c = +1$ — the one innocent-looking bad choice — has its zero at -1, fixed by inversion, so the collapse of Lemma 2 *survives* for $X + 1$: $(\zeta + 1)^{p^{d/2}} = \zeta^{-1} + 1 = \zeta^{-1}(\zeta + 1)$, whence $\operatorname{ord}(X + 1) \mid r(p^{d/2} - 1)$ for even d; moreover $\zeta + 1 = (\zeta^2 - 1)/(\zeta - 1)$ is multiplicatively dependent on the Galois orbit of $\zeta - 1$, so $\{X - 1, X + 1\}$ adds no rank and no burden: a Popovych-style conjecture with $c = 1$ would be no harder for the mechanism than Agrawal's. Thus $c = 2$ is simply the smallest valid choice ($c = -2$ works equally, with Mersenne norm $2^r - 1$). Numerically (`code/ag5b_constant_check.py` of the repository): at even d the section $X + 1$ collapses in 22/22 samples with the top level never needed, while $c \in \{2, 3, -2, 5\}$ collapse only at the single degenerate point $(r, p) = (13, 5)$ of Section 4.

Proposition 10 (the $X + 1$ variant is mechanism-equivalent to Agrawal's conjecture)**.** *Suppose* $n \equiv p^j \pmod{\operatorname{lcm}(r, \rho_r(p))}$ *and* $p - 1 \mid n - 1$ *(the matching congruence plus the Carmichael condition, as in the original L–P sufficient construction). Then* $T(1, n, r)$ *holds modulo* p *outright. Consequently every counterexample to Agrawal's conjecture produced by the mechanism also defeats the variant conjecture whose hypothesis is the pair* $\{T(-1, n, r), T(1, n, r)\}$*: that variant, though formally stronger-looking, is exactly as resistant as Agrawal's — the second hypothesis adds no constraint under these assumptions. The section* $X + 2$ *generally imposes separate order data, but this is not an exclusion of all matching constructions.*

Proof. At the $X = 1$ component, $(1 + 1)^n = 2^n \equiv 2 \pmod p$ since $\operatorname{ord}_p(2) \mid p - 1 \mid n - 1$. In the Φ_r components, $\zeta + 1 = (\zeta^2 - 1)/(\zeta - 1)$, a unit (its norm is $\Phi_r(-1) = 1$), and both $\zeta - 1$ and $\zeta^2 - 1$

are values of $X-1$ at primitive r-th roots of unity, so their orders divide $\rho_r(p) = \mathrm{lcm}_a \,\mathrm{ord}(\zeta^a - 1)$. Hence $(\zeta^a-1)^n = (\zeta^a-1)^{p^j} = \zeta^{ap^j} - 1$ for $a = 1, 2$, and

$$(\zeta+1)^n \;=\; \frac{(\zeta^2-1)^n}{(\zeta-1)^n} \;=\; \frac{\zeta^{2p^j}-1}{\zeta^{p^j}-1} \;=\; \zeta^{p^j}+1 \;=\; \zeta^n+1,$$

using $n \equiv p^j \pmod r$ in the last step. Measured (`code/ag5b_constant_check.py`, check W): 48/48 synthetic mechanism instances pass $T(1, n, r)$, while 0/48 pass $T(2, n, r)$. □

4 Numerical verification

Historical finite checks were made by a deterministic script (`code/ag5_popovych_check.py` of the repository, with committed JSON output), on 56 sample primes spanning

$$(r, d) \in \{(5,4), (13,4), (41,4), (7,6), (7,3), (11,5), (23,11)\}.$$

Frobenius matching for $X+2$ (56/56); the first section's collapse bound fails to absorb $X+2$ at even d (35/36); the top level $e = d$ is present in $\rho'_r(p)$ (55/56); large primes of all levels $e \mid d$ divide $\rho'_r(p)$ (47/47, ≈ 0.28 misses expected under the $1/\ell$ model); $\gcd(r, 2^r+1) = 1$ at all tested r. The single exception in the second and third checks is $r = 13$, $p = 5$, where $\Phi_4(5) = 26 = 2 \cdot 13$ contains no prime outside $2r$ at all — a vacuous top level at one tiny prime, consistent with (indeed predicted by) the generic model. It is nonetheless an accidental finite-field collapse and does not contradict the characteristic-zero statement. These samples do not establish the support hypotheses or asymptotic cost estimates.

5 Scope

Three boundaries. (i) *Mechanism scope*: the results quantify the Frobenius-matching family containing the original sufficient construction; "accidental" solutions of the component equations outside it are not excluded, and no way to construct them is known. (ii) *Conditionality*: the burden *upper* structure (the collapse identities and support bounds) is unconditional; the *lower* bounds use generic sharpness (Definition 6), an Artin-type hypothesis — a construction could target primes whose section orders collapse further; no asymptotic density or economic comparison with smooth-prime pools is proved here. (iii) *Direction*: the conditional support-model comparison is not a universal algorithmic obstruction, an exclusion of every matching construction, or a proof of either conjecture.

References

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